

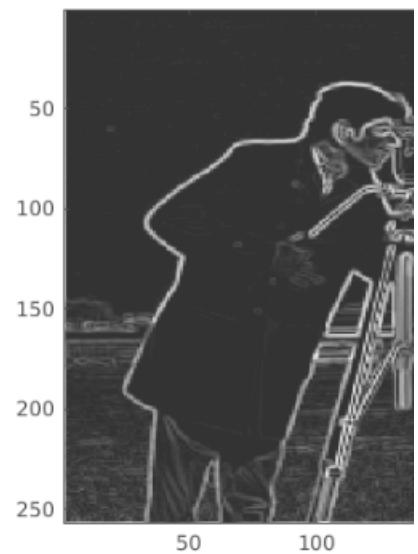
Edge strength

Sobel filters are useful for image edge detection. Horizontal and vertical Sobel filters provide positive filter response for similarly oriented negative response for similarly oriented dark-to-bright edge transitions. Often, we only want to know the magnitude of the edge strength for each pixel.

Write a function called **SobelMagnitude**, which takes as input an image **img**, and outputs **edge_mag**, the magnitude of Sobel filter response by computing the square root of the sum of the squared horizontal and vertical filter response results.

When running the example code, your output should look identical to this:

Output



Function ?

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[MATLAB Documentation \(https://www.mathworks.com/help/\)](https://www.mathworks.com/help/)

```
1 function edge_mag = SobelMagnitude(img)
2 % Use fspecial to generate the horizontal and vertical Sobel filters (sh and sv)
3 sh = fspecial('sobel');
4 sv = sh';
5 % Apply both filters to the image to generate img_sh and img_sv
6 img_sh = conv2(img, sh, 'same');
7 img_sv = conv2(img, sv, 'same');
8 % Compute edge_mag as the square root of the sum of the squared img_sh and img_sv results
9 edge_mag = sqrt(img_sh.^2 + img_sv.^2);
10 end
```

Code to call your function ?

[Reset](#)

```
1 img = imread('cameraman.tif');
2 edge_mag = SobelMagnitude(img);
3 imagesc(img); colormap(gray);
4 figure; imagesc(edge_mag); colormap(gray);
```

[▶ Run Function](#)

Assessment: All Tests Passed

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✔ Is output y correct for random valued input?

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